

10 (**short**) Figure 8 shows scanning electron micrographs of fracture surfaces obtained from metallic samples tested in tension.

(a) For each of the micrographs, briefly describe the main features observed. Comment on the relative magnitude of the energy expended during fracture in each case. Sketch the *nominal stress* – *nominal strain* curves for each case. Account for the main difference between the shapes of the two curves. [5]

(b) If the micrographs were obtained from specimens of the same steel, suggest a possible explanation for the observed differences. Explain your reasoning. Using sketches to illustrate your answer, explain what would be the macroscopic shape of the samples after failure in each case. [5]

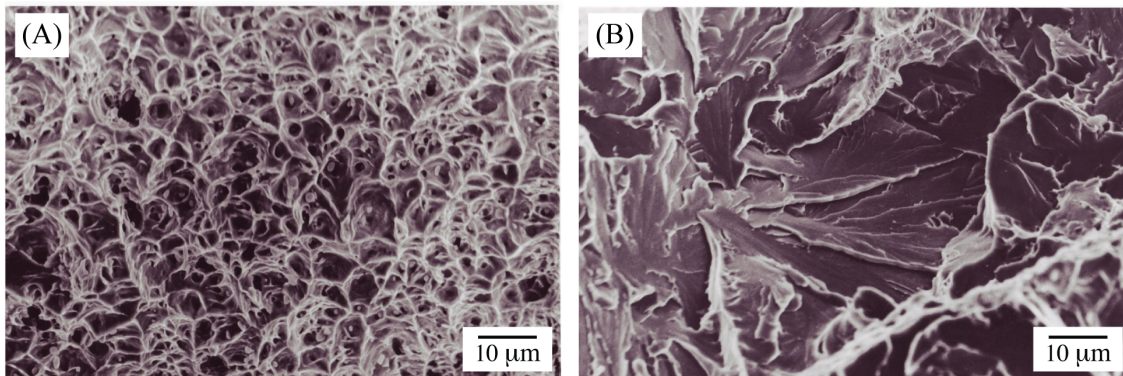


Fig. 8